

Tangent Planes and Linear Approximations

Faruk Uygul

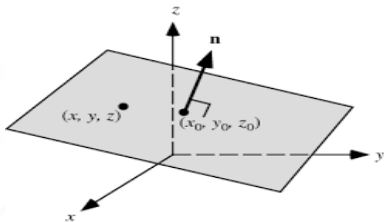
University of Alberta

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Let us begin this section By remembering some facts ABOUT planes and lines in R^3 .

Planes The equation of a plane passing through a point $P(x_0, y_0, z_0)$ with the normal vector $n = \langle a, b, c \rangle$ can be written as

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

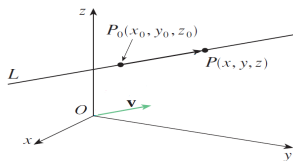


Proof

In $\mathbb{R}^3$, a straight line L in space is determined By:

- A point on L
- The direction of L

Consider the straight line below passing through a point $P_0(x_0, y_0, z_0)$ in the direction of a vector $v = \langle a, b, c \rangle$.



Proposition The equation of L can be written in the following two forms.

▶ $x = x_0 + at, y = y_0 + bt, z = z_0 + ct$
(Parametric Equation of L)

▶ $\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$
(Symmetric Equations L)

Consider a smooth surface S and a point $P(x_0, y_0, z_0)$ on it. Let

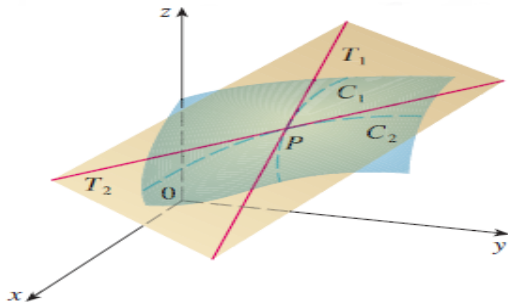
C_1 = intersection of S with the plane $y = y_0$

C_2 = intersection of S with the plane $x = x_0$

T_1 = tangent line to C_1 at the point P

T_2 = tangent line to C_2 at the point P

The tangent plane to S at P is a plane that contains both T_1 and T_2



Proposition Suppose that $z = f(x, y)$ has continuous partial derivatives. An equation of the tangent plane to the surface $z = f(x, y)$ at a point $P(x_0, y_0, z_0)$ is given By

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

The following fact is **very useful** in finding equations of tangent planes.

Let S be a surface given by the equation $F(x, y, z) = 0$. If $P(a, b, c)$ is a point on S , then the vector

$$\vec{n} = \langle F_x(P), F_y(P), F_z(P) \rangle$$

is orthogonal to S at the point $P(a, b, c)$.

Example 1. Find an equation of the tangent plane to

○ $z = e^{x-y}$ at the point $(2, 2, 1)$

○ $z = \frac{x}{y^2}$ at the point $(-4, 2, -1)$

Example 2. Find an equation of the tangent plane to the surface given by the equation

○ $x = y^2 + z^2 - 2$ at the point $A(-1, 1, 0)$

Example 3. Consider the curve C that results from the intersection of the surface $4x^2 + 2y + z^2 = 16$ and the plane $y = 2$.

- Find the parametric equation of the tangent line to C at the point $(1, 2, \sqrt{8})$
- Find the equations of the normal line and the tangent plane to the surface $4x^2 + 2y + z^2 = 16$ at $(0, 6, 2)$.

Recall that the linear approximation to a function $y = f(x)$ is

$$L(x) = f(a) + f'(a)(x - a)$$

$L(x)$: An equation of the tangent line to $y = f(x)$ at $x = a$.

Linear Approximations

If $z = f(x, y)$ is a smooth function, then the **linear approximation** (or the **tangent plane approximation**) at the point $(a, b, f(a, b))$ is given by

$$L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Remark 1. $L(x, y)$ is the equation of the tangent plane at the point $P(a, b, f(a, b))$.

Definition If $z = f(x, y)$, then f is called **differentiable** if

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y$$

where

$$\epsilon_1, \epsilon_2 \rightarrow 0 \text{ as } \Delta x \rightarrow 0 \text{ and } \Delta y \rightarrow 0$$

Therefore, if $f(x, y)$ is differentiable then

$$(x, y) \sim (a, b) \implies f(x, y) \sim L(x, y)$$

There is an alternative definition of differentiability which is somewhat easier to understand.

Definition If a function $f(x, y)$ is differentiable at a point, then

- f is continuous at (a, b)

- $$\lim_{(x,y) \rightarrow (a,b)} \frac{f(x, y) - L(x, y)}{\sqrt{(x - a)^2 + (y - b)^2}} = 0.$$

Therefore, if f is differentiable at $(a, b) \implies$ the partial derivatives at (a, b) must exist.

The following theorem is sufficient but not necessary for f to be differentiable at a point.

Theorem If the partial derivatives f_x and f_y exist at and around (a, b) then f is differentiable at (a, b) .

Example. Show that $f(x, y) = \sqrt{x^2 + y^2}$ is not differentiable at $(0, 0)$.

Solution.

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{h^2}}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h} = DNE$$

Similarly, $f_y(0, 0)$ does not exist either. Therefore, f cannot be differentiable at $(0, 0)$.

Example. Show that for the function

$$f(x, y) = \begin{cases} \frac{7x^2y}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Both $f_x(0, 0)$ and $f_y(0, 0)$ exist, yet f is not differentiable at $(0, 0)$.

Solution.

Note that f is continuous at $(0, 0)$. Now,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

Similarly,

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

Thus, we have $L(x, y) = f(0, 0) + f_x(0, 0)(x - 0) + f_y(0, 0)(y - 0) = 0$.

Now, we need to check if $\lim_{(x, y) \rightarrow (a, b)} \frac{f(x, y) - L(x, y)}{\sqrt{(x - a)^2 + (y - b)^2}} = 0$ is satisfied or not.

$$\begin{aligned}
 \lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y) - L(x,y)}{\sqrt{x^2 + y^2}} &= \lim_{(x,y) \rightarrow (0,0)} \frac{\frac{7x^2y}{x^2+y^2}}{\sqrt{x^2 + y^2}} \\
 &= \lim_{(x,y) \rightarrow (0,0)} \frac{7x^2y}{(x^2 + y^2)^{3/2}}
 \end{aligned}$$

Along the path $y = mx$, we obtain

$$\lim_{x \rightarrow 0} \frac{7x^2(mx)}{(x^2(1+m^2))^{3/2}} = \lim_{x \rightarrow 0} \frac{7x^3m}{x^3((1+m^2))^{3/2}} = \frac{7m}{((1+m^2))^{3/2}}$$

Therefore, the limit DNE. Hence, f is not differentiable at $(0,0)$.

Example. Show that for the function

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

is differentiable at $(0, 0)$.

Solution.

Note that f is continuous at $(0, 0)$.

As in the previous example, $f_x(0, 0) = f_y(0, 0) = f(0, 0) = 0$. Thus, $L(x, y) = 0$.

Now, we need to check if $\lim_{(x,y) \rightarrow (a,b)} \frac{f(x, y) - L(x, y)}{\sqrt{(x-a)^2 + (y-b)^2}} = 0$ is satisfied or not.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - L(x, y)}{\sqrt{x^2 + y^2}} = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{(x^2 + y^2)^{3/2}} \quad (*)$$

$$0 \leq \frac{x^2 y^2}{(x^2 + y^2)^{3/2}} \leq \frac{x^2}{x^2 + y^2} \frac{\sqrt{y^2}}{\sqrt{x^2 + y^2}} \sqrt{y^2} \leq |y|$$

Thus, by the squeeze theorem, $(*)$ is zero.

Remark The red and the blue fractions above are both ≤ 1 .

Example 2 Is the given function differentiable at the indicated point? Find the linearization at that point.

- $f(x, y) = \sqrt{x + e^{4y}}$ at the point $(3, 0)$

- $f(x, y) = \sin(2x + 3y)$ at the point $(-3, 2)$

Example 3. Suppose that f is a differentiable function such that

$$f(2, 5) = 6, \quad f_x(2, 5) = 1, \quad f_y(2, 5) = -1.$$

Use linear approximations to estimate $f(2.2, 4.9)$.

Example 4. Find the linearization to $f(x, y) = \sqrt{x^2 + y^2}$ at the point $(3, 4)$. Use this linearization to estimate $f(3.04, 3.98)$. What is the **error** in this estimation?

Definition If $z = f(x, y)$ then we define the differential (or total differential) of z to be

$$dz = f_x dx + f_y dy = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

Similarly, if $w = f(x, y, z)$ then

$$dw = f_x dx + f_y dy + f_z dz$$

Example 5. Find the differentials of

- $u = x^2 y^3 - x + 3y$

- $w = \frac{7y-3x}{z}$

Example 6. The dimensions of a closed rectangular box are measured as 80 cm, 60 cm and 50 cm, respectively, with a possible error of 0.2 cm in each dimension. Use differentials to estimate the maximum error in calculating the surface area of the box.

Example 7. Use differentials to estimate the value of

$$3.(2.1)^2 - 2.(2.1)(-3.05) + (-3.05)^2$$

Example 8. Use differentials to estimate the maximum error in the calculated volume of a closed cylindrical can that is 10 cm high and 4 cm in diameter if the metal in the top and bottom is 0.1 cm thick and the metal in the sides is 0.05 cm thick.

Example 9.

Consider $z = f(x, y) = x^2 + 3xy - y^2$

1. Find dz
2. If x changes from 2 to 2.05 and y changes from 3 to 2.96 find Δz and dz
3. Compare Δz and dz

Example 10. Find the linear approximation to the function $f(x, y) = \sqrt{20 - x^2 - 7y^2}$ at $(2, 1)$. Use it to approximate $f(1.95, 1.08)$. What is the error in this estimation?